AGRICULTURAL OUTPUT BY STATE

A Data Analytics Capstone Project

Crop Production Analysis for Nigeria's 37 States, 2014–2024

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Table of Contents

1. Executive Summary

This project delivers an interactive Power BI dashboard, "Agricultural Output by State," built to help planners understand how crop production in Nigeria has evolved across all 37 states (36 states plus the Federal Capital Territory) between 2014 and 2024. It covers the country's four major staple crops: Rice, Cassava, Yam, and Maize.

The work followed a full analytics pipeline: a raw dataset was profiled and cleaned in Python, validated against internal consistency checks, modeled into a star schema in Power BI, and visualized across three report pages using DAX-driven measures. Every cleaning decision is documented and traceable, and every number in the dashboard can be reproduced from the published scripts and logs.

The analysis found that Cassava and Yam dominate national output by volume, that production is geographically concentrated in a small number of states, and — the most planning-relevant finding — that national production grew roughly 32% over the decade while average yield per hectare barely moved. This indicates the growth came mainly from cultivating more land rather than farming existing land more efficiently, a distinction with direct implications for where investment should go next.

2. Problem Statement & Objectives

2.1 Problem Statement

Crop production planning and food security decisions in Nigeria require reliable, state-level output data. In practice, this kind of data is often fragmented, inconsistently labeled across sources, or contains gaps and errors by the time it reaches an analyst. This project set out to answer a specific question: which states and zones are driving national crop production, how has yield efficiency changed over time, and where are the clearest opportunities for planning and investment — land expansion versus productivity improvement?

2.2 Objectives

Primary Objective: Build an interactive Power BI dashboard showing state-level crop production trends across Nigeria from 2014 to 2024.

- Clean and validate the raw production dataset so every figure in the dashboard is traceable and defensible.
- Quantify which states and geopolitical zones lead national production, and by how much.
- Distinguish states whose output is driven by land area from states driven by yield efficiency, to support differentiated planning decisions.

3. Data Source & Overview

The project began with a single raw CSV file, raw_crop_output_nigeria_2014_2024.csv, containing 1,145 records at state × crop × year granularity.

Field	Description
state	Nigerian state name
crop	Crop type: Rice, Cassava, Yam, or Maize
year	Reporting year, 2014–2024
area_harvested_ha	Land area harvested, in hectares
yield_t_per_ha	Yield, in tonnes per hectare
production_tonnes	Total production, in tonnes

Coverage: 37 states (36 states + Abuja FCT) × 4 crops × 11 years (2014–2024).

4. Methodology

The project followed a four-stage pipeline, shown below.

Raw CSV → Python Cleaning & Validation → Power Query Modeling (Star Schema) → DAX Measures → Power BI Dashboard

4.1 Data Cleaning (Python)

The raw dataset was profiled and cleaned using pandas. Four issues were identified and resolved, all logged row-by-row in data_quality_log.csv for full traceability:

- Inconsistent state naming: “Abuja FCT” and “FCT Abuja” were both present as separate values for the same entity. All 15 affected rows were standardized to “Abuja FCT.”
- One exact duplicate row was identified and removed.
- 15 rows had a missing production_tonnes value despite having valid area and yield figures; these were recalculated as area_harvested_ha × yield_t_per_ha.
- One severe outlier (Kwara / Yam / 2014) showed a production figure roughly 55× higher than area × yield implied — consistent with a data entry error (e.g., an extra digit). It was recalculated using the same area × yield method.

The cleaned dataset was reduced from 1,145 to 1,144 rows (one duplicate removed), with zero missing values remaining across all fields.

4.2 Data Modeling (Power Query / Power BI)

The cleaned dataset was loaded into Power BI and shaped into a star schema with one fact table and two dimension tables:

- Fact_CropOutput — the core fact table (state, crop, year, area, yield, production).
- Dim_State — a 37-row lookup table mapping each state to one of Nigeria's six geopolitical zones (North Central, North East, North West, South East, South South, South West), enabling zone-level analysis.
- Dim_Year — an 11-row lookup table (2014–2024), carrying both a numeric Year field for time-intelligence calculations and a text YearText field to key the relationship to the fact table.

Both dimension tables were related to the fact table as one-to-many, single-direction relationships, forming a clean star schema.

4.3 DAX Measures

A dedicated _Measures table was created to house the following calculations:

Measure	Definition
Total Production	SUM of production_tonnes
Total Area Harvested	SUM of area_harvested_ha
Avg Yield (t/ha)	Total Production ÷ Total Area Harvested (weighted, not a simple average)
YoY Growth %	Change in Total Production versus the prior year
CAGR 2014–2024	Compound annual growth rate of production over the full period
% of National Production	A state's Total Production ÷ national Total Production

Note: Avg Yield (t/ha) is a weighted average — Total Production divided by Total Area — rather than a simple average of the yield column. This prevents small-area states from skewing the efficiency figure, which matters for planning-grade analysis.

5. Dashboard Design

The final deliverable is a three-page interactive Power BI dashboard titled “Agricultural Output by State.” Each page has a dedicated subtitle reflecting exactly what it analyzes:

Page	Subtitle	What It Shows
Overview	National Overview: Production, Land Use & Yield Across Nigeria's 37 States (2014–2024)	KPI cards (Total Production, Total Area Harvested, Avg Yield, CAGR), a state map, a Top 10 producing states chart, and a crop-share donut chart.
Trends Over Time	Production & Yield Trends by Crop, 2014–2024	Line charts tracking production and yield per crop over the 11-year period, plus a YoY growth column chart.
State Comparison / Planning View	State & Zone Comparison: Ranking Output and Identifying Land- vs. Yield-Driven Growth	A Zone → State × Crop matrix with conditional formatting, a ranked % of national production chart, and a scatter chart plotting area harvested against yield to separate land-driven from efficiency-driven states.

Slicers for Year and Crop are synced across all three pages, a consistent green/earth-tone theme is applied throughout, and crop colors are held constant across every visual so a viewer can track a given crop visually from page to page.

6. Key Insights

Insight 1 — Cassava and Yam dominate national output; Rice lags far behind. Cassava (≈650M tonnes) and Yam (≈569M tonnes) together account for the large majority of cumulative production across the period, while Rice (≈83M tonnes) is the smallest of the four crops by a wide margin.

Insight 2 — Production is geographically concentrated. The top 5 producing states (Benue, Kogi, Kaduna, Oyo, Taraba) account for roughly 37% of total national production across all crops and years. Benue alone leads by a wide margin — roughly double the second-place state.

Insight 3 — National output grew, but efficiency was nearly flat. National production rose about 32% from 2014 to 2024, but the national weighted average yield barely moved (7.27 → 7.29 t/ha). This suggests growth was driven primarily by expanding area harvested rather than by improvements in productivity per hectare — the central finding the State Comparison scatter chart is built to expose at the state level.

7. Recommendations

Priority	Recommendation	Based On
High	Prioritize yield-improvement programs (inputs, extension services, irrigation) over pure land expansion, since national yield has stagnated despite production growth.	Insight 3
Medium	Investigate Rice production specifically — its low volume relative to Cassava/Yam is worth a dedicated deep-dive against national rice demand and import figures.	Insight 1
Low	Use the State Comparison scatter chart to identify high-area/low-yield states as first candidates for productivity investment.	Insight 2, 3

8. Assumptions & Limitations

8.1 Assumptions

- The source dataset's state and crop labels (aside from the FCT naming inconsistency) were treated as accurate and were not independently verified against an external source.
- Where production_tonnes was missing or a clear outlier, area_harvested_ha × yield_t_per_ha was assumed to be the more reliable figure, and the original value was overwritten.

8.2 Limitations

- The dataset covers only 4 crops; conclusions do not extend to other staples (e.g., sorghum, millet, groundnut) that also matter for Nigeria's food security.
- Data is at state-year granularity — no seasonal, LGA-level, or smallholder-vs-commercial breakdown is available, which limits how targeted the recommendations can be.
- 528 state-crop-year combinations present in a full cross-join were absent from the source data (e.g., not every state reports every crop). These were treated as “not grown / not reported” rather than imputed as zero, but this was not independently confirmed against the data source.
- No SQL analysis layer has been built yet for this project; the work completed to date is Python (cleaning) + Power BI/DAX (modeling and visualization) only.

9. Future Enhancements

- Add a SQL layer to answer specific planning questions directly with queries — e.g., which states are under-producing relative to available land.
- Add a Dim_Crop table with crop metadata (e.g., staple vs. cash crop) if the analysis expands beyond simple crop-name slicing.
- Validate the 528 missing state-crop-year combinations against the original data source to confirm they represent true non-cultivation rather than reporting gaps.
- Extend the dataset to additional staple crops if source data becomes available.

10. Conclusion

The “Agricultural Output by State” project turned an inconsistent, error-prone raw dataset into a validated, well-documented analytical asset and a purpose-built dashboard for crop planning conversations. Beyond the visuals, the project produced a reusable, auditable pipeline — a cleaning script, a data quality log, a data dictionary, and a modeled star schema — so every number in the final dashboard can be traced back to a documented decision. The clearest strategic takeaway, that Nigeria's crop production growth over the last decade has come from more land rather than more efficient land, gives planners a concrete, evidence-based starting point for where to focus investment next.

11. Project Deliverables

Deliverable	Description
Cleaned dataset	crop_output_nigeria_2014_2024_clean.csv — 1,144 validated rows
Cleaning script	01_clean_data.py — reproducible Python cleaning pipeline
Data dictionary	data_dictionary.csv — field-level documentation for all tables and measures
Data quality log	data_quality_log.csv — row-level record of every change made during cleaning
Power BI dashboard	“Agricultural Output by State” — 3-page interactive .pbix
Project README	Repository documentation following the project's standard template
This report	Full project report, start to finish

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